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DEPARTMENT OF ELECTRONICS AND COMMUNICATION ENGINEERING
EVENT-TRIGGERED OUTPUT FEEDBACK MODEL PREDICTIVE CONTROL
FOR PATH FOLLOWING OF AUTONOMOUS VEHICLES
Major Project Phase II Review-2

BY

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ABSTRACT

- ▶ Attacks on the sensor-controller (S-C) channel, which enable the infusion of false data, present substantial risks to the security of autonomous vehicles.
- ▶ This work endeavors to address the challenge of guaranteeing secure path-following control for autonomous vehicles under the threat of False Data Injection (FDI) attacks, which through the utilization of an event-triggered output feedback model predictive control approach.
- ▶ To ensure the controller receives uncompromised system states amidst FDI attacks, a distributed robust multivariate observer (DRMO) is employed, which can facilitate the estimation and differentiation of uncompromised system states and attack signal simultaneously.
- ▶ Building upon the uncompromised system states provided by the observer and considering various constraint conditions, a predictive controller is formulated to ensure secure control of autonomous vehicle path following.
- ▶ Additionally, an event-triggered mechanism is introduced, dynamically adjusting the controller update frequency, resulting in significant savings in computational and communication resources.

INTRODUCTION

- ▶ Autonomous vehicles need a control system to follow the road or path correctly. Normal controllers keep updating at every moment, even when it is not needed, which wastes time and resources.
- ▶ Event-triggered control updates only when there is an error or the vehicle starts to go off the path.
- ▶ Output feedback helps use sensor data when all information is not directly available.
- ▶ It makes Model Predictive Control (MPC) is used to predict the vehicle's movement and decide the best action. the system accurate, efficient, and suitable for real-time driving.

LITERATURE REVIEW

S.NO	TITLE	AUTHOR	YEAR	FOCUS	RESEARCH GAP
1.	Event-Triggered MPC-Preview Control for Vehicle Path Tracking.	Zhang et al.	2024	Combined preview information with event-triggered MPC for path tracking.	Limited to simple models, not tested in real vehicles.
2.	Adaptive Dynamic Programming with Event-Triggered Sampling	Zhang et al.	2024	Combined learning with event-triggered control for vehicle path following.	High complexity; needs practical testing on real car.
3.	Output-Feedback H_∞ Path Tracking for Autonomous Vehicles.	Liu et al.	2023	Used output-feedback control to handle model uncertainty in vehicles.	No event-triggering, so high computation load remains.
4.	Event-Triggered MPC for Autonomous Vehicle with Rear Steering.	SAE Pape	2023	Used event-triggered MPC for vehicle with rear steering to reduce updates.	Assumed full state measurement, no output feedback.
5.	Robust MPC for Autonomous Vehicle Path Following	Wang et al	2023	Used robust MPC for stable path following	No event-triggered strategy, more control updates

S.NO	TITLE	AUTHOR	YEAR	FOCUS	RESEARCH GAP
6.	Learning-Based MPC for Autonomous Driving	Zhang et al.	2022	Combined MPC with machine learning for tracking	Combined MPC with machine learning for tracking
7.	Event-Triggered Control for Autonomous Vehicle Systems	Li et al.	2022	Used event-triggered control to reduce communication.	Did not combine with MPC for path tracking
8.	Distributed MPC for Vehicle Path Following	Zhao et al.	2021	Used distributed MPC for multi-vehicle tracking	No event-triggered mechanism, high computation
9.	Adaptive MPC for Autonomous Vehicle Steering	Kumar et al.	2021	Adaptive MPC for steering control	Only simulation results, no real vehicle test
10.	Event-Based Path Tracking for Intelligent Vehicles	Park et al.	2021	Event-based control for vehicle path tracking	Limited robustness under disturbances

PROBLEMS IDENTIFIED

1. High Computational Complexity

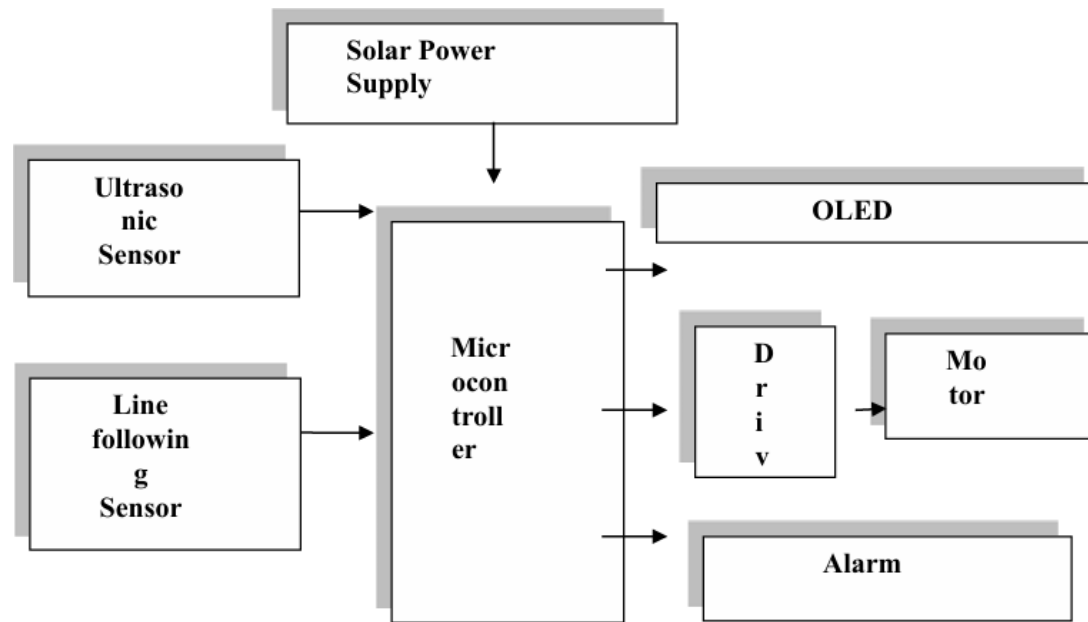
Description: Most existing path-tracking methods (especially MPC and learning-based approaches) require heavy computation and frequent controller updates. This makes real-time implementation difficult on vehicles with limited onboard hardware.

2. Lack of Practical Real-Time Validation

Description: Many studies are validated only through simulations and assume full vehicle state information. Very few methods are tested on real vehicles, so their performance in real-world conditions (noise, disturbances, road changes) is still uncertain.

ARCHITECTURE OF PROPOSED METHODOLOGY

BLOCK DIAGRAM :-



HARDWARE REQUIREMENTS

- Microcontroller
- Ultrasonic sensor
- IR Sensor
- Driver Circuit
- Motor
- OLED
- Buzzer

SOFTWARE REQUIREMENTS

- Embedded C
- Arduino IDE

ADVANTAGES

1. Reduced Computational Load
 - Processor doesn't work all the time
2. Saves Energy
 - Works only when it is needed
3. Maintains Accuracy
 - Still keeps the vehicle on the correct path
4. Improves System Life Span
 - Steering and breaks are not used continuously
5. Good For Real Time System
 - Works well in real vehicles

DISADVANTAGES

1. Design Is Complex
 - Hard to design compared to normal control methods
2. Risk Of Delay
 - Sometimes correction may come late
3. May Reduce Accuracy
 - Vehicle may slightly deviate from path
4. Stability Issues
 - System can become unstable(unsafe)
5. Safety Concerns In Critical Conditions
 - Vehicle may not react quickly

CONCLUSION

- Event-triggered output feedback MPC is a smart and efficient control method for autonomous vehicles. It reduces unnecessary calculations and actions by updating control only when needed, which saves energy, time, and computing power.

REFERENCES

- ▶ Zhang J. et al. (2024). Event-triggered MPC-preview control for vehicle path tracking.
- ▶ Liu Q. et al. (2023). Output-feedback H_∞ path tracking for autonomous vehicles.
- ▶ GuoS. et al. (2023). Self-triggered robust output feedback MPC.
- ▶ Zhang H. et al. (2024). Adaptive dynamic programming with event-triggered sampling for vehicles.
- ▶ SAE Paper (2022). Event-triggered MPC for autonomous vehicle with rear steering

Thank You!

